

# Notes de cours

$$\vec{k} = \frac{2\pi}{\lambda} \vec{u} \quad \text{ici } \vec{u}_z = \vec{u}$$

$$\vec{u}_z \begin{vmatrix} 0 \\ 0 \\ 1 \end{vmatrix} \quad \text{car } \vec{u}_z = 0\vec{u}_x + 0\vec{u}_y + 1\vec{u}_z$$

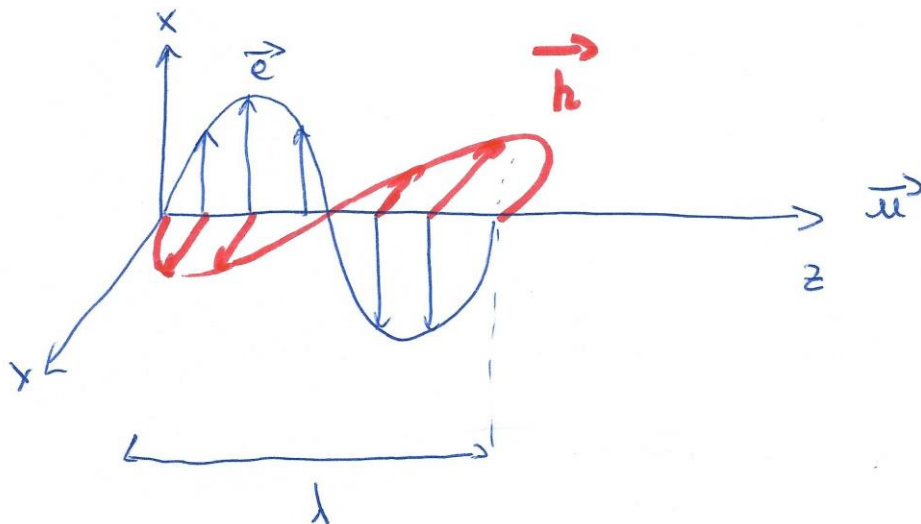
dans la base  $(0, \vec{u}_x, \vec{u}_y, \vec{u}_z)$

Rappel: soit  $\vec{a} \begin{vmatrix} a_x \\ a_y \\ a_z \end{vmatrix} \quad \left( \Leftrightarrow \vec{a} = a_x \vec{u}_x + a_y \vec{u}_y + a_z \vec{u}_z \right)$

et  $\vec{b} \begin{vmatrix} b_x \\ b_y \\ b_z \end{vmatrix} \quad \left\{ \begin{array}{l} \vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z \\ \hat{=} \\ \text{produit scalaire} \end{array} \right.$

$$\text{Ici } \vec{k} \cdot \vec{OM} = \frac{2\pi}{\lambda} \vec{u} \cdot \vec{OM} = \frac{2\pi}{\lambda} \vec{u}_z \cdot \vec{OM}$$

$$= \frac{2\pi}{\lambda} \begin{vmatrix} 0 \\ 0 \\ 1 \end{vmatrix} \cdot \begin{vmatrix} x \\ y \\ z \end{vmatrix} = \frac{2\pi}{\lambda} z$$



## Produit vectoriel :

soit  $\vec{a}$  de coordonnées  $(a_x, a_y, a_z) \Leftrightarrow \vec{a} \begin{vmatrix} a_x \\ a_y \\ a_z \end{vmatrix}$

$\vec{b} \quad \text{---} \quad (b_x, b_y, b_z) \Leftrightarrow \vec{b} \begin{vmatrix} b_x \\ b_y \\ b_z \end{vmatrix}$

$$\vec{a} \wedge \vec{b} = \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = \begin{vmatrix} a_y b_z - a_z b_y \\ b_x a_z - b_z a_x \\ a_x b_y - a_y b_x \end{vmatrix}$$

↑  
produit  
vectoriel

↑  
c'est un vecteur.

Sur la figure 2  $\vec{e}(M, t) = \vec{E}_0 \cos(\omega t - \frac{2\pi}{\lambda} z)$

ici  $\vec{e}$  vibre selon  $(Ox) \Rightarrow \vec{E}_0 = E_0 \vec{u}_x$ .

$$\Rightarrow \vec{e}(M, t) = \boxed{E_0 \cos(\omega t - \frac{2\pi}{\lambda} z)} \vec{u}_x = \begin{vmatrix} e_x = E_0 \cos(\omega t - \frac{2\pi}{\lambda} z) \\ e_y = 0 \\ e_z = 0 \end{vmatrix}$$

**$e_x$**

Calculons  $\frac{1}{z_0} \vec{u}_z \wedge \vec{e} = \frac{1}{z_0} \begin{vmatrix} 0 \\ 0 \\ 1 \end{vmatrix} \wedge \begin{vmatrix} E_0 \cos(\omega t - \frac{2\pi}{\lambda} z) \\ 0 \\ 0 \end{vmatrix}$

↑  
 $\vec{u}_z$

$$= \frac{1}{z_0} \begin{vmatrix} 0 \\ E_0 \cos(\omega t - \frac{2\pi}{\lambda} z) \\ 0 \end{vmatrix}$$

$$\Rightarrow \frac{1}{z_0} \vec{u}_z \wedge \vec{e} = \frac{1}{z_0} E_0 \cos(\omega t - \frac{2\pi}{\lambda} z) \vec{u}_y = \vec{h}$$

$\Rightarrow \vec{h}$  ~~est~~ vibre bien selon  $Oy$  comme sur la figure 2.